

HP PageWide 556, 586, 765, 774, 776, 779, 780, 785, E55650, E58650, E75160, P75250, P77440, E77650, E77660, P77940, P77950, P77960 - Print Quality Troubleshooting Guide

ish_4885409-4869728-16 | For HP And Channel Partner Internal Use

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Use this topic to diagnose and troubleshoot print-quality problems.

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Printers

Learn about printers supported by this guide.

This document applies to following mid-range small engine printers

- HP PageWide Pro 750
- HP PageWide Pro 772
- HP PageWide Pro 777
- HP PageWide Managed P55250dw
- HP PageWide Managed P57750dw MFP
- HP PageWide Managed P75040
- HP PageWide Managed P75050
- HP PageWide Managed P75060
- HP PageWide Managed P77740 MFP
- HP PageWide Managed P77750 MFP
- HP PageWide Managed P77760 MFP

NOTE:

When diagnosing print quality issues, always verify that the customer is using genuine HP supplies. The supply pages will help to verify that. If non genuine HP supplies are in use, ask the customer to replace them with genuine HP supplies prior to performing any other Troubleshooting steps. HP **cannot** guarantee print quality while using Non HP supplies.

Introduction

Learn about general print-quality troubleshooting guidelines.

Paper

A3 (Ledger) printers paper have wider paper paths than the A4 (Letter) printers.

A3 paper size is always loaded in portrait orientation in the trays for A3 printers.

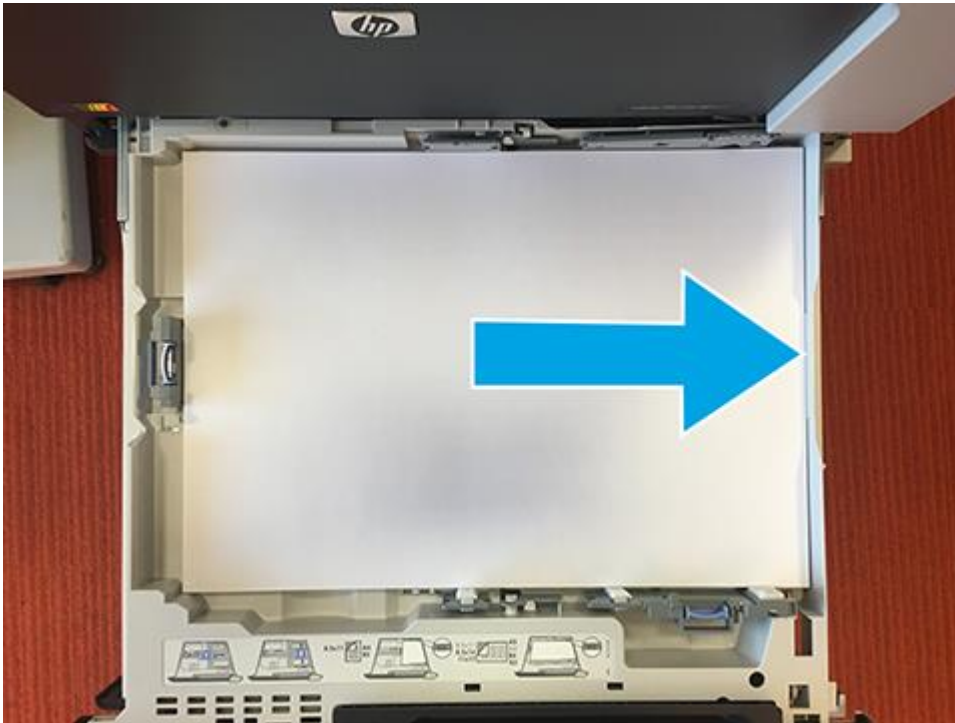
A4/Letter paper size can be loaded in portrait or landscape orientations.

It is very important to know how the customer is loading the paper into the trays in order to correctly diagnose PQ issues, especially for repetitive defects.

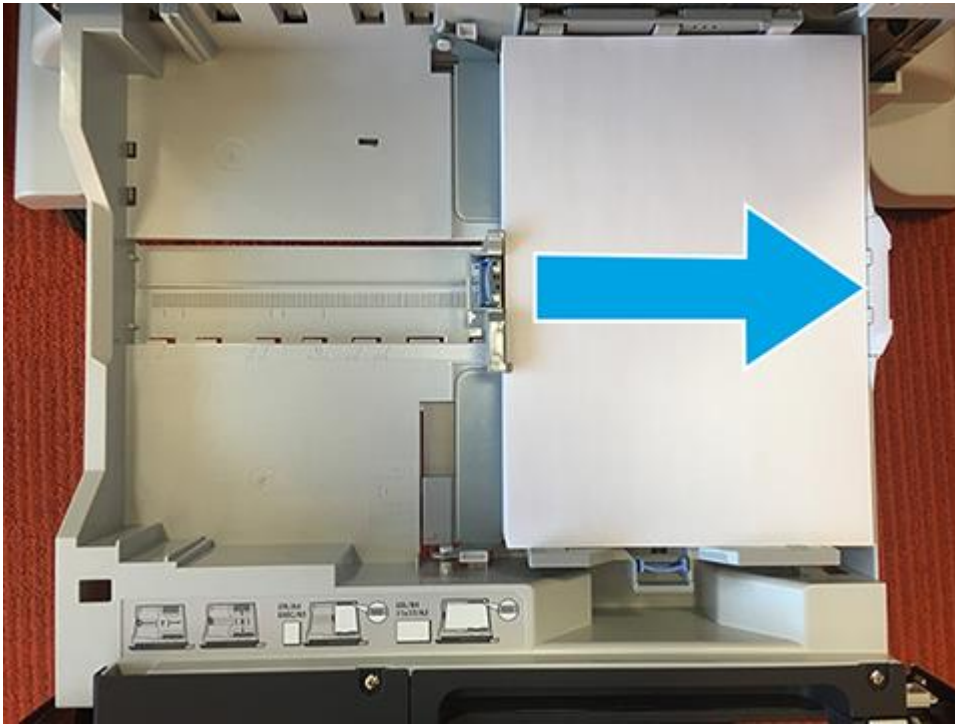
Instead of asking the customer about vertical or horizontal paper orientation, which can be confusing, ask the customer to send you the configuration page. On the page you will see the correct tray settings (A3, A4R, etc.). Use the table below to instruct the customer to view the paper in portrait or landscape mode.

- For A3 paper, have the customer inspect the PQ defects in portrait mode.
- For A4 paper, ask the customer if they loaded the paper as A4 or A4R.
 - A4 - have the customer inspect the PQ defects in landscape mode.
 - A4R - have the customer inspect the PQ defects in portrait mode.
- This is due to how the paper travels along the printer's paper path, as seen in the following images.

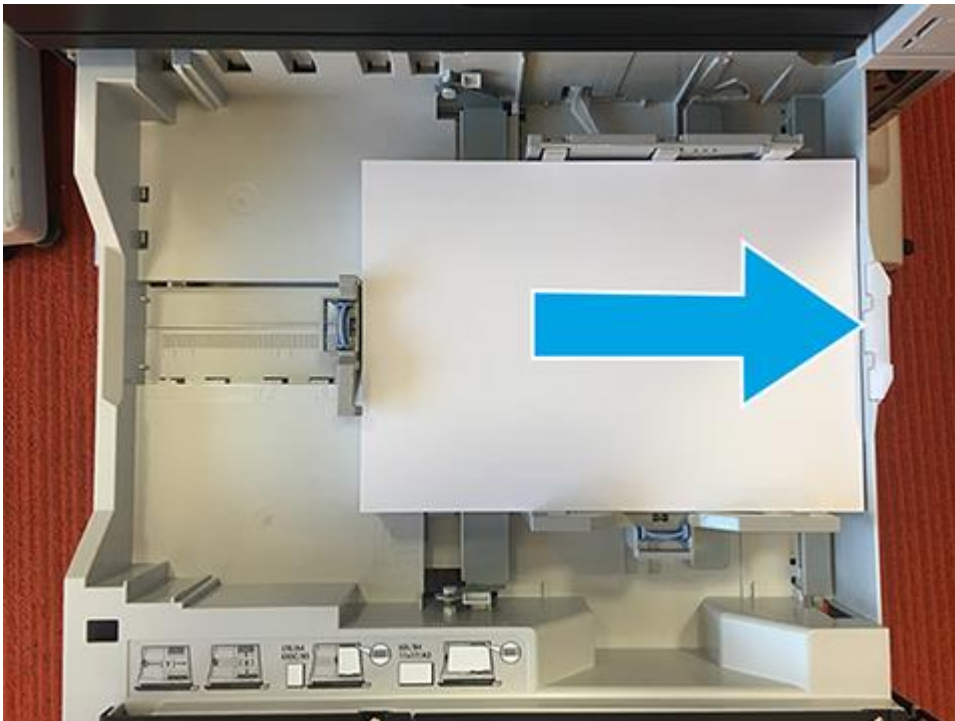
A3 MOTION IS IN PORTRAIT MODE



A4 MOTION IS IN LANDSCAPE MODE.



A4R MOTION IS IN PORTRAIT MODE.



Paper being used

The paper type and size being used is also important. Make sure that the paper meets the PageWide printer specifications for the specific printer, and that the tray settings match it.

For example, if a tray is loaded with labels and paper type is set to normal the fuser temperature might not be enough to correctly fuse the toner to the paper, causing a PQ issue.

Environment

Most HP PageWide printers have environmental sensors installed that adjust the fuser temperature to ensure good print quality. However, poor print quality might occur if a printer is located in a place where the temperature and/or humidity are outside of the recommended specifications.

Example; High humidity usually causes paper curl, and even changing some of the settings might not solve the problem.

Ask the customer about the printer location. If needed, request the customer to move the unit to a normal office environment (within the recommended specifications for that printer - see the printer service manual).

Information to collect

- Supply Status page to get information about toner cartridges and maintenance kits.
- Configuration page to check tray settings.
- Usage page to get information about how the printer is being used regarding paper sizes.

Printer-specific image defects quick reference

Use this topic as a reference to print-quality troubleshooting topics included in the guide.

The table below provides a visual reference of printer image defects and symptoms. A link is provided to access the advanced troubleshooting information for each defect.

NOTE:

Go to for a list of printers supported in this guide.

TIP:

For basic print-quality troubleshooting information, Search for: HP PageWide Enterprise Color, HP PageWide Managed Color - Resolving color print quality problems (doc id - c05805661).

IMAGE DEFECTS QUICK REFERENCE

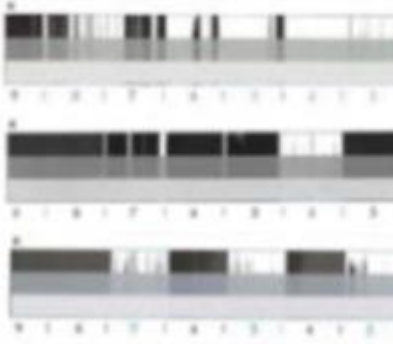
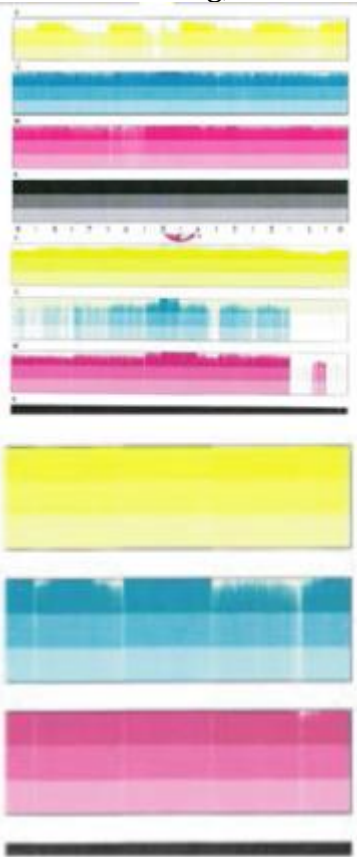
Image defect	Image defect	Image defect
Vertical bands For advanced troubleshooting, see: 	White streaks across one or all colors For advanced troubleshooting, see: 	Streaks improve down the page For advanced troubleshooting, see: 

IMAGE DEFECTS QUICK REFERENCE

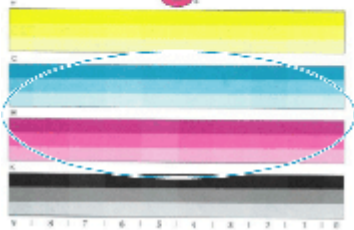
Image defect	Image defect	Image defect
Streaks across the entire page – single or multiple colors consistent down the page For advanced troubleshooting, see: 	Color Mix For advanced troubleshooting, see: . 	Color variation across bar (Waterfall) For advanced troubleshooting, see: . 
Color intensity variation across bars For advanced troubleshooting, see: <u>Print Quality Report:</u> 2019010401010000000000000000000000 • Examine the color bars below. If the bars show significant vertical or horizontal streaks, check the printhead, but also check the media. • If printout is missing data not in the picture, get more data at the end of the page. • 10% maximum is allowed for ink and color and paper. 	Bars not even, columns of misaligned print, or vertical dark lines For advanced troubleshooting, see:	Streaks (yellow only) For advanced troubleshooting, see:). 

IMAGE DEFECTS QUICK REFERENCE

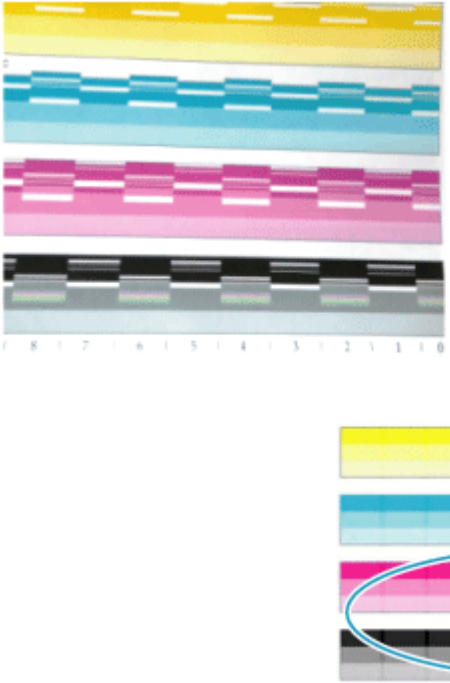
Image defect	Image defect	Image defect
		
In-line streak(s) and drip(s) For advanced troubleshooting, see:	Large sections of wrong or missing color For advanced troubleshooting, see:	Vertical white lines For advanced troubleshooting, see:

IMAGE DEFECTS QUICK REFERENCE

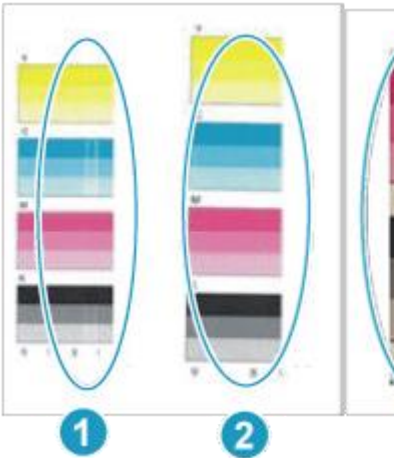
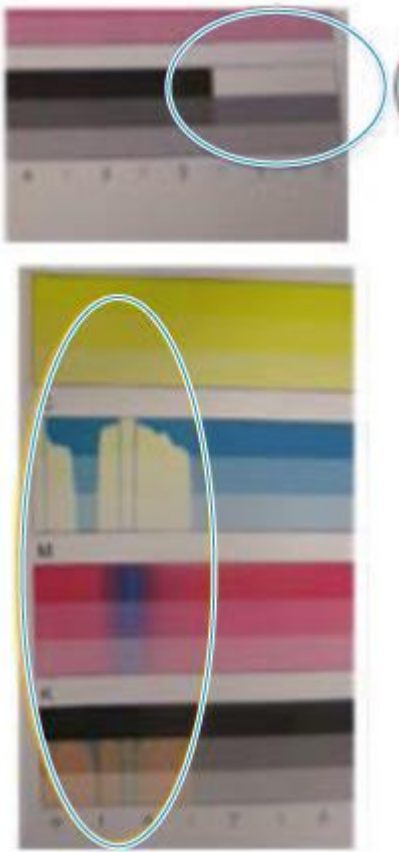
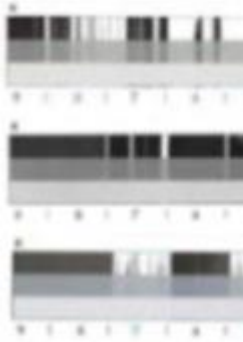
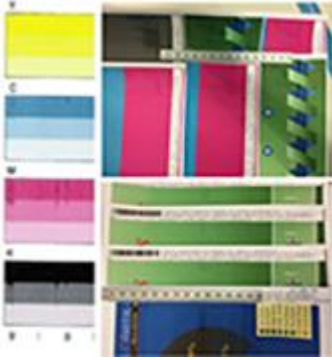
Image defect	Image defect	Image defect
		
Ink smear / Redeposit For advanced troubleshooting, see: .		

IMAGE DEFECTS QUICK REFERENCE

Image defect	Image defect	Image defect
		

Go here for [Printer pre-checks](#), [Engine Diagnostic pages](#), and [cleaning](#)

Advanced print-quality troubleshooting

Learn about specific-image defect advanced print-quality troubleshooting.

TIP:

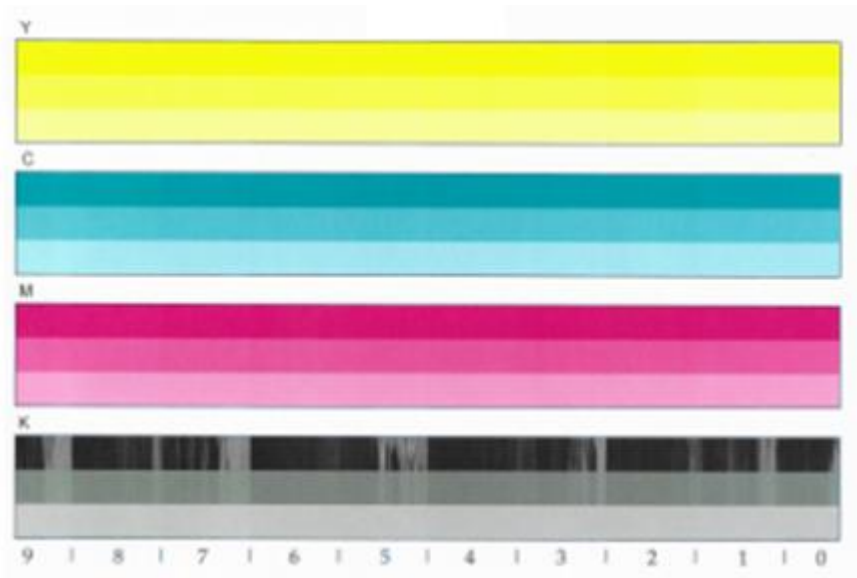
For basic print-quality troubleshooting information, Search for: HP PageWide Enterprise Color, HP PageWide Managed Color - Resolving color print quality problems (doc id - c05805661).

The following troubleshooting and help topics can be found at the end of print quality issues:

- [Printer pre-checks](#)
- [Check the print settings](#)
- [Printer driver considerations](#)
-
-
-

Black Ink Fading

Learn about print-quality troubleshooting black ink faded print.



Description

Learn about how the image defect appears on the page.

Black appears to be fading on the page.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

This condition happens on the printer after long periods of time without printing.

Solution

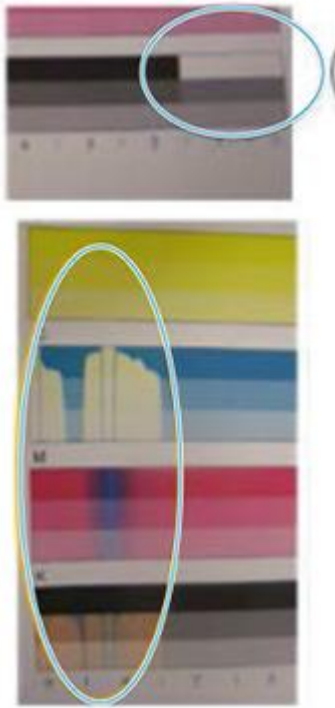
Learn about solutions for resolving the image defect.

1. Print a few more full blacks pages to help clear up the issue.
2. If printing full black pages does not resolve the issue, replace the Mech kit..

Large sections of wrong or missing color

Learn about print-quality troubleshooting Large sections of wrong or missing color.

Wrong or missing color (large)



Description

Learn about how the image defect appears on the page.

Large area(s) of wrong or missing color(s) on the page.

NOTE:

Colors might be present at top of page, but go away by bottom of page.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

This defect is most likely caused by air inside the printhead.

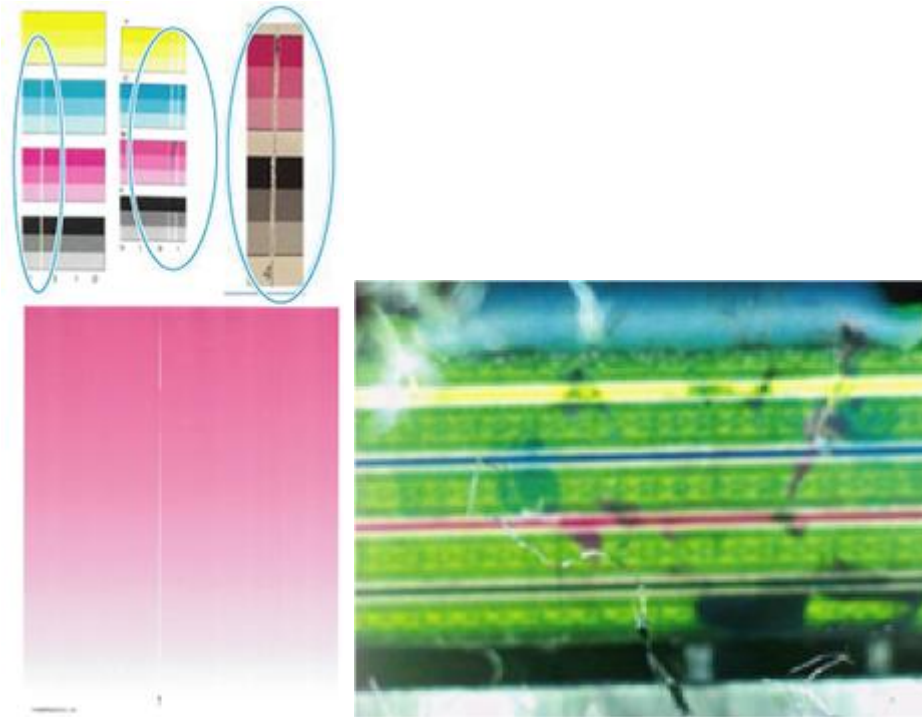
Solution

Learn about solutions for resolving the image defect.

1. Make sure genuine HP ink is installed.
2. Print a print-quality report.
3. Perform a printhead cleaning level 2 procedure.
4. Print a print-quality report, and then compare it to the first report printed.
5. If the error persists, install a print mechanism (PM) kit.

Small sections of wrong or missing color

Learn about print-quality troubleshooting Small sections of wrong or missing color.



Description

Learn about how the image defect appears on the page.

Shim whisker (fibers stuck onto the shim under the printbar).

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

This defect is most likely caused by air inside the printbar.

Solution

Learn about solutions for resolving the image defect.

1. Make sure genuine HP ink is installed.
2. Print a print-quality report.
3. Perform a printhead cleaning level 2 procedure.
4. Print a print-quality report, and then compare it to the first report printed.

NOTE:

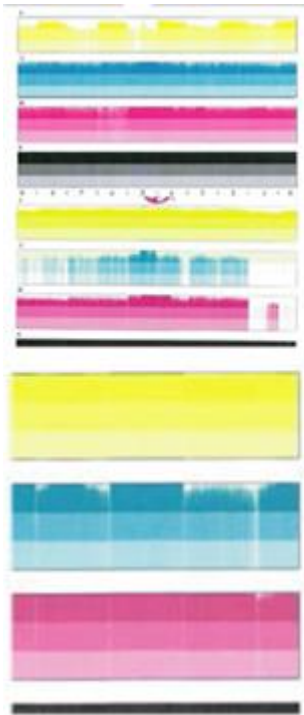
If the report shows improved print quality, perform the cleaning level 2 procedure again (do not exceed three times).

5. If print quality did not improve, order the advanced cleaning kit. Follow the instructions provided in the kit.
Part number: (CN459-67006)
6. If the error persists, install a print mechanism (PM) kit.

Streaks that get better down the page.

Learn about print-quality troubleshooting streaks on pages.

STREAKS THAT GET BETTER DOWN THE PAGE.



Description

Learn about how the image defect appears on the page.

Streaks across entire page (multiple colors improving down the page).

NOTE:

The defect appears worse in colors (black being fairly robust).

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

This defect is most likely caused by the print bar being left uncapped for an extended time.

Solution

Learn about solutions for resolving the image defect.

1. Make sure genuine HP ink is installed.
2. Print a print-quality report.
3. Perform a printhead cleaning level 2 procedure.
4. Print a print-quality report, and then compare it to the first report printed.

NOTE:

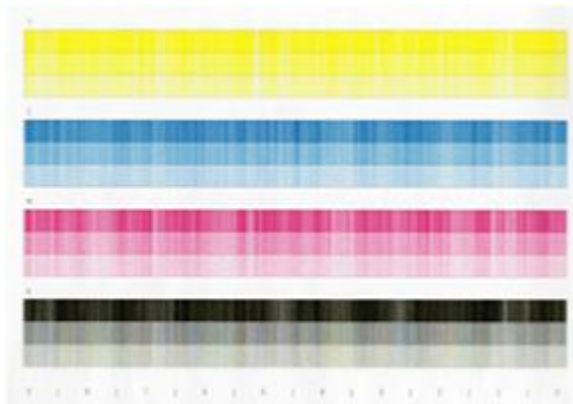
If the report shows improved print quality, perform the cleaning level 2 procedure again (do not exceed three times).

5. If the error persists, install a print mechanism (PM) kit.

Streaks consistently down the page

Learn about print-quality troubleshooting streaks on pages.

STREAKS CONSISTENTLY DOWN THE PAGE



Description

Learn about how the image defect appears on the page.

Streaks across entire page (multiple colors consistent down the page).

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

If the printer recovers after sitting for 20 minutes, the web wipe used to clean the nozzles might have reached its end of life.

If the issue persists (after the printer has sat for 20 minutes), the defect is most likely caused by a mechanical failure (for example, the service sled is not rotating).

Solution

Learn about solutions for resolving the image defect.

1. Make sure genuine HP ink is installed.
2. Print a print-quality report.
3. Perform a printhead cleaning level 2 procedure.
4. Print a print-quality report, and then compare it to the first report printed.

NOTE:

If the report does not show improved print quality, wait 20 minutes and then print another print-quality report.

- If the report shows improved print quality (after waiting the prescribed 20 minutes), check the engine page count.
 - If the page count is seventy-five thousand or greater, the error might return. Replace the service sled
5. If the error persists and the page count is seventy-five thousand or less, install a print mechanism (PM) kit.

Streaks (yellow only)

Learn about print-quality troubleshooting Streaks (yellow only).



Description

Learn about how the image defect appears on the page.

Streaks appear in the yellow color plane only.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

TWINE: build up around the endcap region on print bar.

Solution

Learn about solutions for resolving the image defect.

1. Make sure genuine HP ink is installed.
2. Print a print-quality report.
3. Perform a printhead cleaning level 2 procedure.
4. Print a print-quality report, and then compare it to the first report printed.

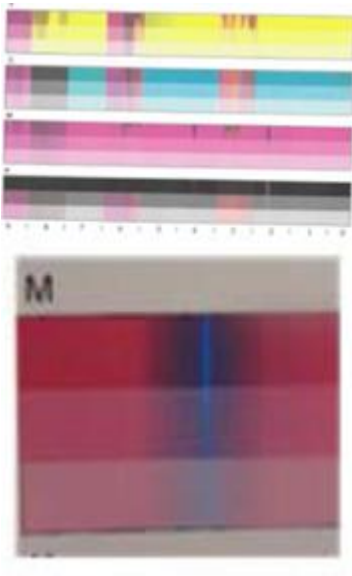
NOTE:

If the report shows improved print quality, perform the cleaning level 2 procedure again (do not exceed three times).

5. Print a print-quality report. If the error persists, install a print mechanism (PM) kit.

Color mix

Learn about print-quality troubleshooting Color mix defects.



Description

Learn about how the image defect appears on the page.

Color bleeds on the page.

Color mixing might be localized (not the entire die) or appear across an entire die.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

Color mixing might be caused by either an internal or external contamination source. External contamination clear-up with continued printing, or printhead cleaning. Internal contamination might require replacing the print bar.

Solution

Learn about solutions for resolving the image defect.

1. Make sure genuine HP ink is installed.
2. Print a print-quality report.
3. Perform a printhead cleaning level 2 procedure.
4. Print a print-quality report, and then compare it to the first report printed.

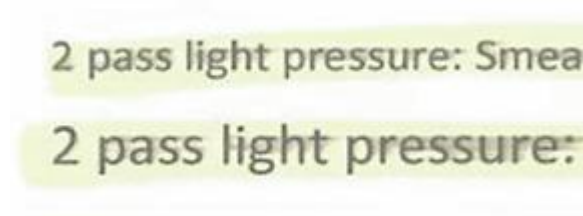
NOTE:

If the report shows improved print quality, perform the cleaning level 2 procedure again (do not exceed three times).

5. If the error persists, install a print mechanism (PM) kit.

Highlighter smear

Learn about print-quality troubleshooting Highlighter smear.



Description

Learn about how the image defect appears on the page.

This defect appears as smeared text when using a highlighter.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

When using a highlighter pen on printed pages, a smear might be seen (this defect is caused by both the media type and the highlighter pen).

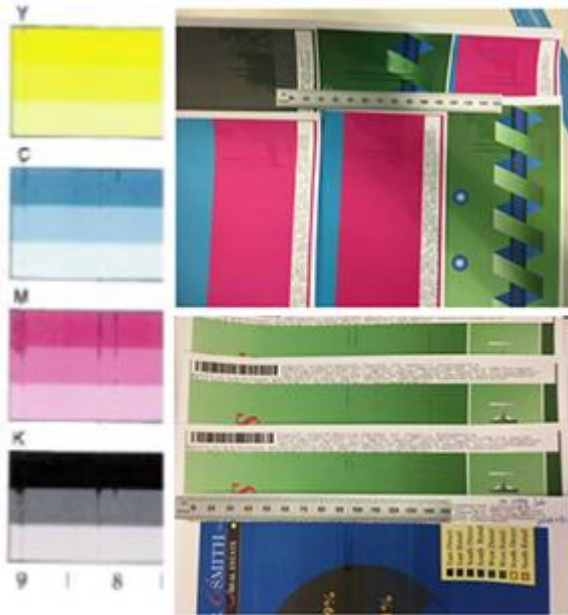
Solution

Learn about solutions for resolving the image defect.

1. Try different brand of highlighter.
2. When using a highlight pen, use a single pass without too much pressure.
3. Try using a different media type.

Ink smear

Learn about print-quality troubleshooting image placement defects or skew.



Description

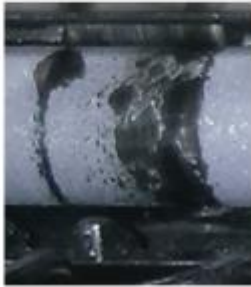
Learn about how the image defect appears on the page.

This defect appears as smeared ink.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

- Decreased distance between the paper and the printhead nozzles due to contamination
- Build up along paper path



Solution

Learn about solutions for resolving the image defect.

1. Make sure genuine HP ink is installed.
2. Print a print-quality report.
3. Open the left door, remove the duplexer (ICU), and then lower the flap (push the green tab).
4. Check the shaft. If ink buildup is present, try using the grit roller cleaning kit. Other buildup mitigation methods:
 - Open the ink slider, and reduce the print speed (Dry Time) setting.
 - Open the ink slider, and reduce the ink saturation (Saturation) setting.
 - Open the ink slider, try different combinations of print speed and saturation (Dry Time and Saturation) settings.
 - Try using a different paper type.
5. If the error persists, install a print mechanism (PM) kit.

Image stretch

Learn about print-quality troubleshooting image stretch defects.



Description

Learn about how the image defect appears on the page.

The image is stretched vertically or skewed (or both).

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

Grit roller contamination causes image stretch or skew issue.

If the ink contamination on the grit roller is evenly distributed, only a stretched image is created.

If ink contamination on grit roller is unevenly distributed across the width, then image skew and stretch is created.

Solution

Learn about solutions for resolving the image defect.

1. Make sure genuine HP ink is installed.
2. Print a print-quality report.
3. Open the left door, remove the duplexer (ICU), and then lower the flap (push the green tab).
4. Check the shaft. If ink buildup is present, try using the grit roller cleaning kit.

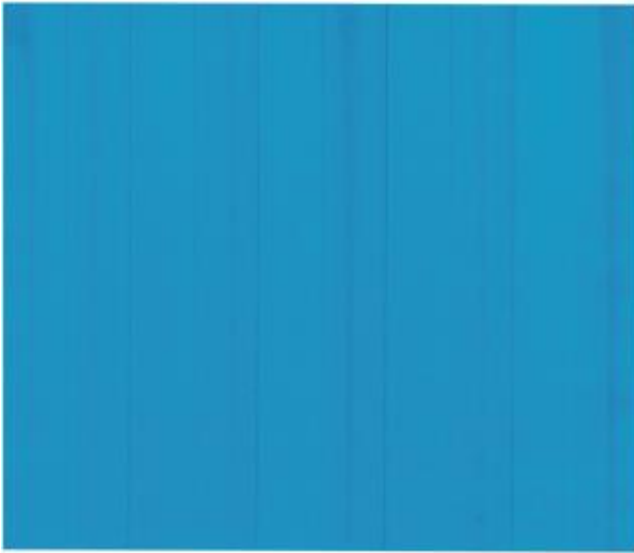
Other buildup mitigation methods:

- Open the ink slider, and reduce the print speed (Dry Time) setting.
- Open the ink slider, and reduce the ink saturation (Saturation) setting. .

- Open the ink slider, try different combinations of print speed and saturation (Dry Time and Saturation) settings.
 - Try using a different paper type
5. If the error persists, install a print mechanism (PM) kit.

Waterfall lines

Learn about print-quality troubleshooting waterfall lines.



Description

Learn about how the image defect appears on the page.

Streaks and band appear on the first few pages.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

If printer has been idle for an extended time, the first couple pages might have a defect that resembles a waterfall down the page.

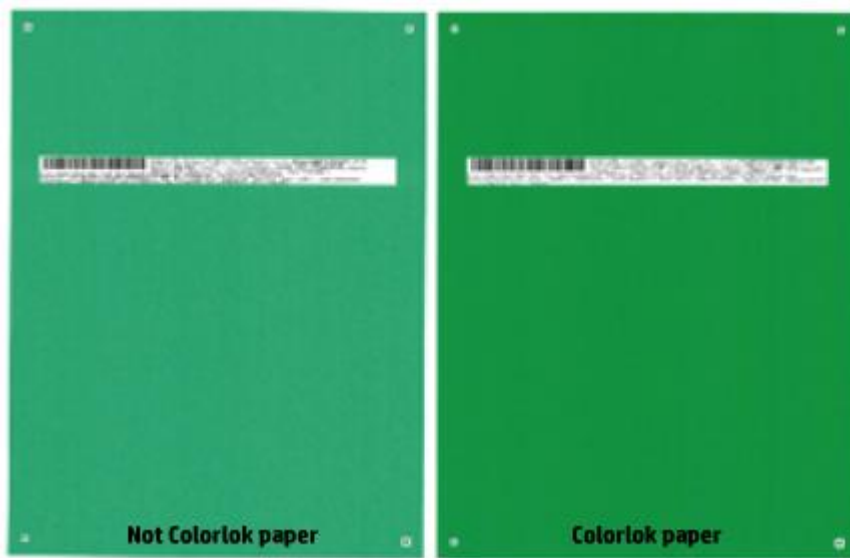
Solution

Learn about solutions for resolving the image defect.

1. This error self recovers after several pages print.
2. If not recoverable, perform the cleaning level 1 procedure (do not exceed four times).

Washout

Learn about print-quality troubleshooting partial washout.



Description

Learn about how the image defect appears on the page.

Colors appear washed out (not vivid).

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

This is expected performance between Colorlok paper versus non ColorLok paper.

Solution

Learn about solutions for resolving the image defect.

1. Use ColorLok paper to for best color performance.

Transition error

Learn about print-quality troubleshooting transition error.



Description

Learn about how the image defect appears on the page.

Image transition defect.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

This defect might appear at the top of the page, at the middle of the page, or at the bottom of the page.

The main cause is page vibration when paper contacts, enters, or leaves a specific roller.

Solution

Learn about solutions for resolving the image defect.

1. Open the ink slider, and reduce the print speed (Dry Time) setting.

Starwheel tracking

Learn about print-quality troubleshooting starwheel tracking.



Description

Learn about how the image defect appears on the page.

Image starwheel tracking defect.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

Ink is being picked up by the star wheels and redeposited on the page.

Solution

Learn about solutions for resolving the image defect.

1. Open the ink slider, try increasing the print speed and decreasing the saturation (Dry Time and Saturation) settings.
2. Try using a different paper type.

Ink not drying

Learn about print-quality troubleshooting ink not drying issues.



Description

Learn about how the image defect appears on the page.

All sorts of exaggerated issues or ink not drying at all.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

Dark lines, missing ink, and other defects are much more visible when using non-inkjet compatible paper.

Solution

Learn about solutions for resolving the image defect.

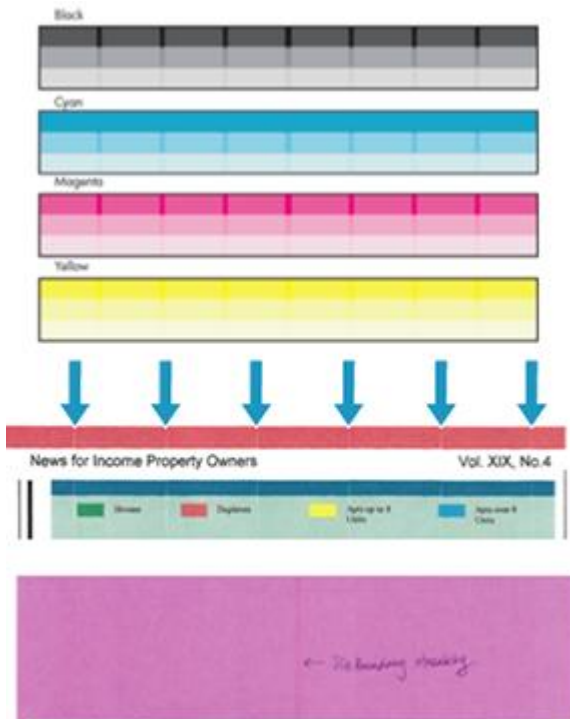
1. Make sure to use InkJet compatible paper.

Pen alignment calibration

Learn about print-quality troubleshooting Pen alignment calibration issues.

NOTE:

The upper most image below is an exaggerated example of this defect. Lines might be dark or white, and might only be on specific regions of the page.



Description

Learn about how the image defect appears on the page.

This defect appears as lines aligned with printhead boundaries.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

Streaks align with die boundary because the printhead is not aligned.

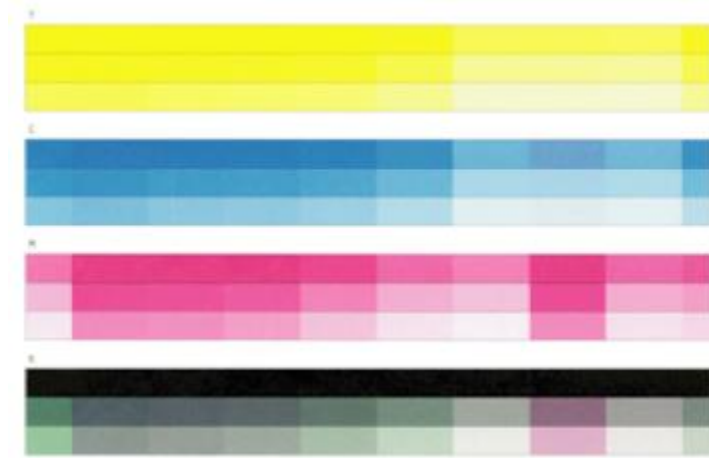
Solution

Learn about solutions for resolving the image defect.

1. From the printer control panel, open the following menus:
 - Device Maintenance
 - Calibration/Cleaning
 - Advance Calibration Support
2. Select Continue, and then select Pen Align to align the printhead.

Color difference

Learn about print-quality troubleshooting color difference issues.



Description

Learn about how the image defect appears on the page.

This defect appears as a color difference between printhead nozzles.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

Density calibration error.

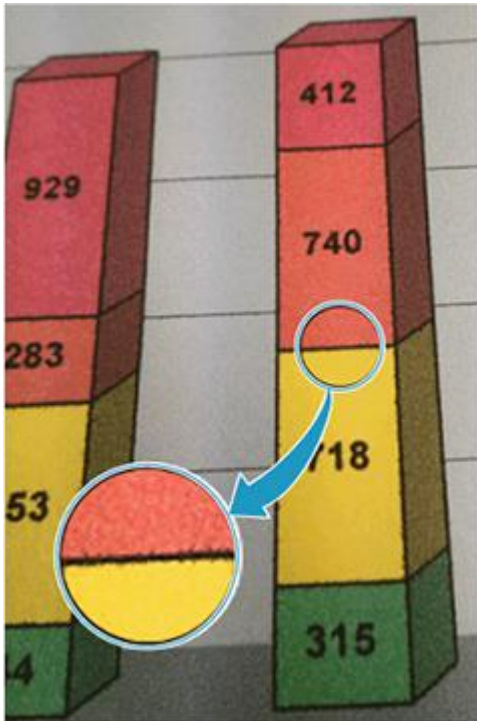
Solution

Learn about solutions for resolving the image defect.

1. From the printer control panel, open the following menus:
 - Device Maintenance
 - Calibration/Cleaning
 - Advance Calibration Support
2. Select Continue, and then select Color Density Calibration to calibrate the printhead.

Blurred boundaries

Learn about print-quality troubleshooting Blurred boundaries issues.



Description

Learn about how the image defect appears on the page.

This defect appears as a blurry or fuzzy boundary between black and color images.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

Ink saturation or paper type error causes this issue.

Solution

Learn about solutions for resolving the image defect.

1. Potential mitigation methods:

- Make sure that HP Colorlok paper is used in the printer.
- Open the ink slider, and reduce the ink saturation (Saturation) setting.
- Open the ink slider, and reduce the ink spread (Black Ink Spread) setting

NOTE:

Use the procedures in this section to resolve most print-quality issues (for example, streaks or fading, missing black or other colors, blurred or fuzzy text, and ink streaks or smears). Try the following solutions in the order

presented. When one of the solutions resolves the issue, there is no need to continue with the other procedures.

Paper curl

Learn about print-quality troubleshooting output curl.



Description

Learn about how the image defect appears on the page.

This defect appears as paper with a wet feel or curl.

Conditions that can cause the defect

Learn about printer conditions that might cause the image defect.

Curl can be seen when printing with high ink coverage, and there are variations between paper.

Solution

Learn about solutions for resolving the image defect.

1. Potential mitigation methods:

- Open the ink slider, and reduce the ink saturation (Saturation) setting.
- Try printing with bigger page margins.
- Try using heavier weight paper.
- Store pages with the printed side down.

Printer pre-checks, Engine Diagnostic pages, and cleaning

Learn about printer Printer pre-checks, Engine Diagnostic pages, and cleaning.

Printer pre-checks

Learn the printer pre-checks and how to use them.

Use the following checks and settings to help diagnose the print quality issue.

Check the control-panel display

Check the control-panel display for messages

Check the control panel to see if it displays any messages that indicate that a supply needs to be replaced or that a supply is in the very low condition. If it does, replace the supply item.

1. Print a supply status page to check the consumables.

Inspect the ink cartridge for damage

Learn how to check the ink cartridges.

Refilled or re-manufactured ink cartridges

HP does not recommend using non-HP supplies, either new or re-manufactured. Because they are not HP products, HP cannot control the design or quality of non-HP supplies. If you are using a refilled or re-manufactured ink cartridge and are not satisfied with the print quality, replace the cartridge with an original HP cartridge.

NOTE:

If you replace a non-HP ink cartridge with an original HP ink cartridge, the printhead will still contain non-HP ink. Until the printhead is depleted of that ink and is supplied with ink from the newly installed original HP ink cartridge, the print defect might remain. Depleting the ink may require printing as many as several thousand pages of normal text.

1. Remove the ink cartridge from the product and verify that there is no debris in the cartridge path.
2. Examine the metal connector of the ink cartridge.

CAUTION:

Do not touch the metal connector that is on the edge of the cartridge. Fingerprints on the metal connector can cause print-quality problems.

NOTE:

If there are scratches or other damage on the metal connector, replace the ink cartridge.

If an ink cartridge must be replaced, look on the cartridge or print the Supplies Status Page to find the part number for the correct original HP ink cartridge.

EXAMINE THE INK CARTRIDGE



3. If the metal connector does not appear to be damaged, push the cartridge gently back into its slot until it locks in place. Print a few pages to see if the problem has resolved.

Verify type of paper in use

Learn what to look for in the paper that is being used.

The type of paper used can contribute to print-quality problems, such as those listed below:

- The printing is too light or seems faded in areas.
 - Specks of ink are on the printed pages.
 - Ink is smearing on the printed pages.
 - Printed characters seem malformed.
 - Printed pages are curled.
1. Check the type of paper used. Always use a paper type and weight that this printer supports. In addition, follow these guidelines when selecting paper:
 - Use paper that is of good quality and free of cuts, nicks, tears, spots, loose particles, dust, wrinkles, voids, staples, and curled or bent edges.
 - Use paper that has not been previously printed on.

- Use paper that is designed for use in inkjet printers.
- Use paper that is not too rough. Using smoother paper generally results in better print quality.
- Check that the paper type is correctly set for the trays in use and that the paper type for the trays matches the paper type selected in the driver.

NOTE:

Use the Tray Configuration dashboard menu to verify or change default paper types for each tray.

2. **NOTE:**

3. Based on HP internal testing using a range of plain papers, HP highly recommends using papers with the ColorLok logo for this product.



4.

5. For more information, go to <http://www.hp.com/go/printpermanence>.

Check the print settings

Learn about the print settings.

Use the following to check different print settings.

Use the printer driver that best meets the print job. Use a different printer driver if the printed page has unexpected lines in graphics, missing text, missing graphics, incorrect formatting, or substituted fonts. The available print drivers are listed in this section.

Print Quality settings

Learn what to set up in print quality settings.

1. Open the print dialog box in the software program being used to send the print job.
2. Select your printer, and then choose Properties. The printer properties dialog box opens.
3. Choose the correct tab to find the settings.
4. Review the following options, and then make changes if necessary.
 - Paper size: Make sure that the selected option matches the selected paper size.

- Paper type: If one of the options matches required paper type exactly, select that option.
 - Print quality: If the print-quality of the print job is unacceptable, increase the print quality. To print more quickly, decrease the print quality.
5. choose OK, and then choose OK again to start the print job.

Print in gray scale settings

Learn how to set up print in gray scale.

1. Open the print dialog box in the software program being used to send the print job.
2. Select your printer, and then choose Properties. The printer properties dialog box opens.
3. Choose the correct tab to find the color settings either under Features or Color.
4. Under Color Options, make sure that Print in grayscale is not selected.
5. choose OK, and then choose OK again to start the print job.

NOTE:

If these steps resolved the issue, further troubleshooting is not required. If the issue persists, continue to Printer driver considerations.

Printer driver considerations

Learn about the print drivers for use with this printer.

Use the printer driver that best meets the print job. Use a different printer driver if the printed page has unexpected lines in graphics, missing text, missing graphics, incorrect formatting, or substituted fonts. The available print drivers are listed in this section.

HP PCL 6 driver

Learn about the HP PCL 6 driver

- This driver is provided as the default driver. This driver is automatically installed when using the printer CD.
- This driver is recommended for all Windows environments.
- This driver provides the overall best speed, print quality, and product-feature support for most users, and was developed to align with the Windows Graphic Device Interface (GDI) for the best speed in Windows environments.

- This driver might not be fully compatible with third-party and custom software programs that are based on PCL 5.
- Download this drive at <http://www.hp.com/go/upd>.

HP UPD PS driver

Learn about the HP UPD PS driver.

- This driver is recommended for printing with Adobe software programs or with other highly graphics-intensive software programs
- This drive provides support for printing from postscript emulation needs, or for postscript flash font support.
- Download this drive at <http://www.hp.com/go/upd>.

HP UPD PCL5

Learn about the HP UPD PCL5.

- This driver is recommended for general office printing in Windows environments.
- This driver is compatible with previous PCL versions and older HP Inkjet printers.
- This driver is the best choice for printing from third-party or custom software programs.
- This drive is the best choice when operating with mixed environments that require the product to be set to PCL 5 (UNIX, Linux, mainframe).
- This driver is designed for use in corporate Windows environments to provide a single driver for use with multiple printer models. This driver is preferred when printing to multiple printer models from a mobile Windows computer.
- Download this drive at <http://www.hp.com/go/upd>.

HP UPD PCL6

Learn about the HP UPD PCL 6 driver.

- This driver is recommended for printing in all Windows environments.
- TThis driver provides the overall best speed, print quality, and product-feature support for most users, and was developed to align with the Windows Graphic Device Interface (GDI) for the best speed in Windows environments.
- This driver might not be fully compatible with third-party and custom software programs that are based on PCL 5.
- Download this drive at <http://www.hp.com/go/upd>.

Printhead status

Learn about how to check printhead status.

When print-quality problems are detected, check the printhead status.

- If non HP ink is in use, the printhead status is: Not Available.
- If the printhead status indicates missing nozzles (poor health), perform a printhead cleaning procedure.

NOTE:

See Advanced Print Quality Tools the **Service mode functions** section of the printer Troubleshooting Manual.

- If the printhead status is OK, print a nozzle health page and check the drop detect item to verify the gauge reading is correct.

NOTE:

See Major functionality for PQ troubleshooting the printer Troubleshooting Manual.

If the gauge reading matches the nozzle health page pattern, other issues might be causing the print quality problem. **Do not** replace the printhead.

Print-quality report (v01)

These steps involve printing a page with color bars to identify print-quality issues.

Print quality report is a useful page which shows different density fill bar chart with four colors. There are numbers below the bar to indicate the printhead numbering, and the line in between indicates the printhead boundary.

NOTE:

Depending on the FW revision, the line to indicate the printhead boundary might not be accurate, it could be off as much as 5 mm for printhead zero. There is a CR 119406 to fix this issue, and the fix has been implemented into version 2307823 for 23r, and has been merged into version 2454949 for 24s.

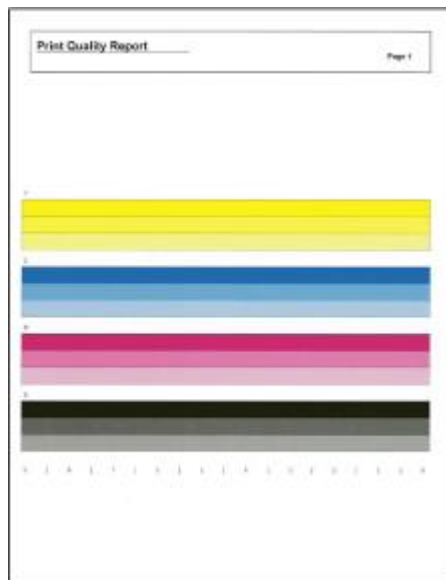
1. At the printer control panel, open the following menus: Administration Troubleshooting Print Quality Pages
2. Select Print Quality Report, and then select Print to print the report.
This page contains four bands of color, which are divided into the groups as indicated in the figure below. By examining each group, you can determine if the problem is due to a particular ink cartridge.

NOTE:

Each bar has three distinct shades, but otherwise the color should vary minimally across the bar. If any major defects appear in one or more color

bars, clean the printhead as explained below. If all the color bars show little or no streaking, go to Resolve Ink Smears.

PRINT QUALITY REPORT



Printhead Cleaning

Use this procedure to clean the printhead.

1. Load plain letter-size or A4-size paper in the tray.
2. At the printer control panel, open the following menus: Device Maintenance Calibration/Cleaning
3. Select Printhead Cleaning, and then select Continue to begin the cleaning process.

There are three levels of cleaning in sequence. At end of each level, the printer will produce a print quality report. If streaks appear in any of the color bars, continue with next level of cleaning.

NOTE:

Number each of the print quality report pages in order to track print-quality improvement.

Resolve Ink Smears

Use the following procedure to clean ink smears.

1. When prompted, load plain letter-size or A4-size paper.
2. At the printer control panel, open the following menus: Device Maintenance Calibration/Cleaning.

3. Select Resolve Ink Smear, and then select Start to begin the cleaning process. A page feeds through the product slowly. Retain this page for further print-quality evaluation.

NOTE:

If the error persists, contact your Global Business Unit (GBU).

4. Under Color Options, make sure that Print in grayscale is not selected.
5. choose OK, and then choose OK again to start the print job.

NOTE:

If these steps resolved the issue, further troubleshooting is not required. If the issue persists, continue to Printer driver considerations.

Print quality reports and diagnostic pages

Learn about Print quality reports and diagnostic pages.

Use these reports and diagnostic pages to verify and resolve the print problems.

Print-quality report (v01)

These steps involve printing a page with color bars to identify print-quality issues.

Print quality report is a useful page which shows different density fill bar chart with four colors. There are numbers below the bar to indicate the printhead numbering, and the line in between indicates the printhead boundary.

NOTE:

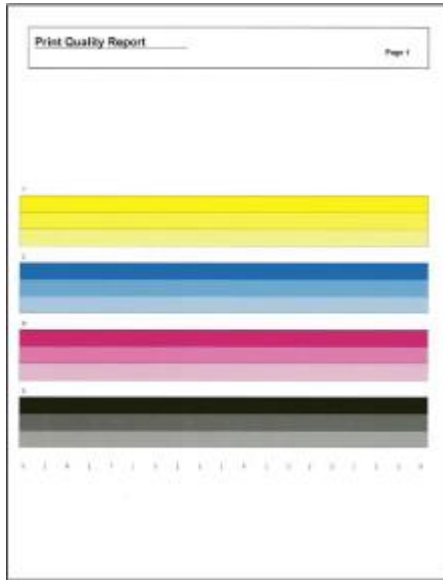
Depending on the FW revision, the line to indicate the printhead boundary might not be accurate, it could be off as much as 5 mm for printhead zero. There is a CR 119406 to fix this issue, and the fix has been implemented into version 2307823 for 23r, and has been merged into version 2454949 for 24s.

1. At the printer control panel, open the following menus: Administration Troubleshooting Print Quality Pages
2. Select Print Quality Report, and then select Print to print the report.
This page contains four bands of color, which are divided into the groups as indicated in the figure below. By examining each group, you can determine if the problem is due to a particular ink cartridge.

NOTE:

Each bar has three distinct shades, but otherwise the color should vary minimally across the bar. If any major defects appear in one or more color bars, clean the printhead as explained below. If all the color bars show little or no streaking, go to Resolve Ink Smears.

PRINT QUALITY REPORT



Print-quality report (v02)

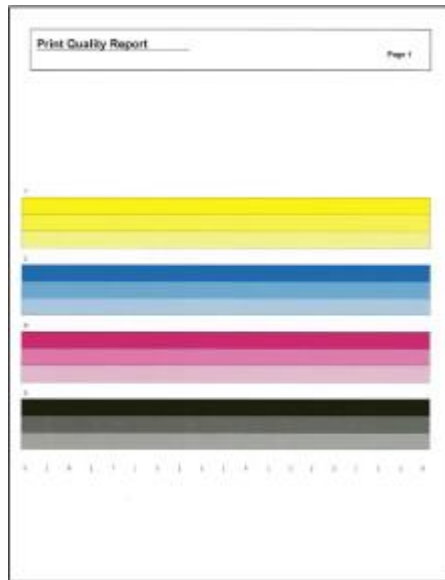
These steps involve printing a page with color bars to identify print-quality issues.

Print quality report is a useful page which shows different density fill bar chart with four colors. There are numbers below the bar to indicate the printhead numbering, and the line in between indicates the printhead boundary.

NOTE:

Depending on the FW revision, the line to indicate the printhead boundary might not be accurate, it could be off as much as 5 mm for printhead zero. There is a CR 119406 to fix this issue, and the fix has been implemented into version 2307823 for 23r, and has been merged into version 2454949 for 24s.

1. At the printer control panel, open the following menus: Administration Troubleshooting Print Quality Pages
2. Select Print Quality Report, and then select Print to print the report.
This page contains four bands of color, which are divided into the groups as indicated in the figure below. By examining each group, you can determine if the problem is due to a particular ink cartridge.
PRINT QUALITY REPORT



Printhead diagnostic page

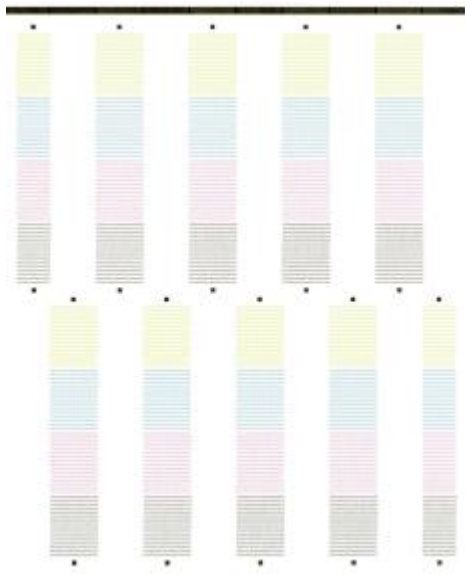
Use this procedure to print the Printhead diagnostic page.

The printhead diagnostic page is a useful page to determine the health of the print bar. There are ten blocks of lines, and each block represents a printhead, and each line corresponds to a nozzle. Check whether the line is missing or being misdirected, to evaluate the health of the print bar.

NOTE:

Depending on the FW revision, the line to indicate the printhead boundary might not be accurate, it could be off as much as 5 mm for printhead zero. There is a CR 119406 to fix this issue, and the fix has been implemented into version 2307823 for 23r, and has been merged into version 2454949 for 24s.

1. At the printer control panel, open the following menus: Administration Troubleshooting Print Quality Pages Advanced Print Quality Pages.
2. Select Continue, and then select Printhead Diagnostic Page to print the page.



Engine diagnostic page

Use the following procedure to print the engine diagnostic pages.

The engine diagnostic page is a useful page to get engine related information. It has information from various portions of the other advanced diagnostic pages.

NOTE:

Some information on this page is encrypted.

- Generic product information (for example, the printer serial number and page counts)
 - Ink delivery system information (for example, the cartridge serial number and whether original HP ink is installed)
 - Print bar related information (for example, printhead register values)
1. At the printer control panel, open the following menus: Administration Troubleshooting Print Quality Pages Advanced Print Quality Pages.
 2. Select Continue, and then select Engine Diagnostic Page to print the page.

NOTE:

Depending on the engine status, a blank page might print with a The product is not able to communicate with all components and cannot retrieve the data for the requested report. Close all doors and trays and try to print the report again message on it. Follow instructions and attempt to print the page again..

ENGINE DIAGNOSTIC PAGE (1 OF 2)

<u>XXXX XXXXX XXXXXXXX</u>				
43. COOOOOO:	Yellow	Magenta	Cyan	Black
44. XXXXXXX XXXXXXX XXXXX:				
45. COOOOOO XXXXX:	Ready	Ready	Ready	Ready
46. COOOOOO XXXXX XXXXX:				
	-1	-1	-1	-1
47. XXXXXXXXXXX XXXXX:	Rx 981A	Rx 981A	Rx 981A	Rx 981A
48. XXXXX XXXXX OX:	0	0	0	0
49. XXXXX XXXXX TX:	0	0	0	0
50. XXX/XXXX XXXXX Bxx:	0	0	0	0
51. XXXX Ux COOOOOO:	0	0	0	0
52. XXXXX XXXXX:	374664925	374673999	374670584	377129891
54. Mxx. XXXX:	09/17/2015	09/17/2015	09/17/2015	10/01/2015
55. COOOOOO XXXXX:	--	--	--	--
56. XXXXXXX XXXX:	09/17/2018	09/17/2018	09/17/2018	10/01/2018
57. TX:	3	3	3	3
58. XXXX:	0	0	0	0
59. XXX:	0	0	0	0
60. XX. 02-850-0000-00-	0000			

1	63	66	62	63	5f	5f	5b	61	5e	69
2	2a	29	2b	2a	2b	2c	2d	2b	2c	29
3	d7	db	c6	d5	cd	d3	c3	c9	c6	dc
4	24	4	4	14	14	14	14	4	4	14
5	45	46	46	45	44	45	44	46	44	46
6	1e	1e	1e	1e	1e	1e	1e	1e	1e	1e
7	c8	c8	c8	c8	c8	c8	c8	c8	c8	c8
8	0	3	0	0	3	0	3	0	2	3

Learn about Printhead cleaning.

There are three automatic printhead cleaning levels (starting with level 1). When a cleaning level is completed a print quality report prints, and then the user can choose to continue to next level or not (depending on what print-quality report shows).

Use these steps to perform a Automatic printhead cleaning.

- Level 1 and level 2 service the printhead to clear clogged nozzles (for example, by wiping the print bar and spitting ink). Level 2 utilizes more wipes to service the print bar than level 1 (and more spitting of ink, especially with colors).
- Level 3 is calibration (including pen alignment calibration and color density calibration).

Automatic printhead cleaning

1. At the printer control panel, open the following menus:
 - Device Maintenance
 - Calibration/Cleaning
 - Printhead cleaning

Manually choose a printhead cleaning level

Use these steps to manually clean the printheads.

1. At the printer control panel, open the following menus:
 - Device maintenance
 - Calibration/Cleaning
 - Advanced Calibration Support
2. Select Continue, and then select from a list of printhead cleaning options.
 - Printhead Cleaning Level 1
 - Printhead Cleaning Level 2
 - Pen Align
 - Color Density
 - Calibration Print Quality Report

Ink Slider

Learn how to use the Ink slide.

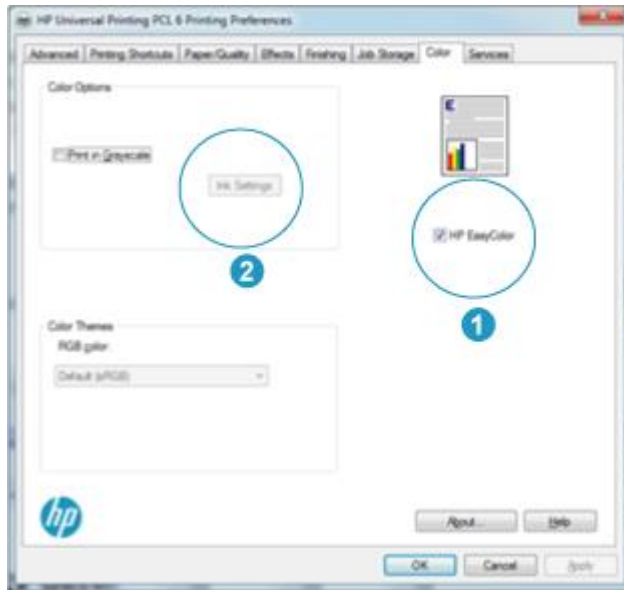
The Ink slider is a printer driver feature. The following drivers support the ink slider feature:

- UPD 6.0.1 (not a public release, special request needed) and 6.1.0 and later
- 586 and 556 discrete driver
- 586 and 556 UPD driver

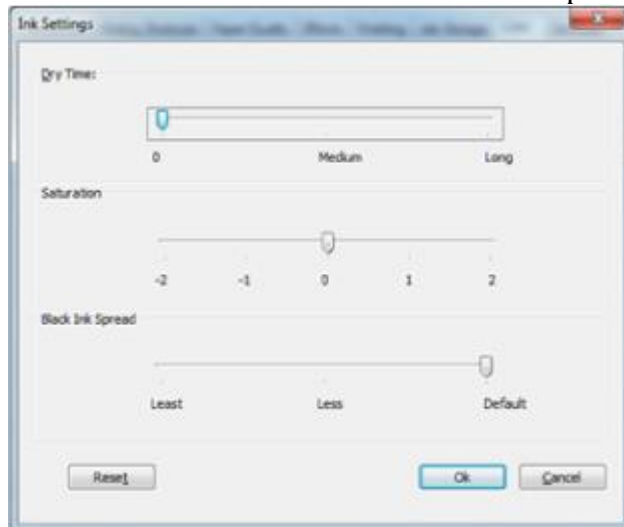
NOTE:

The X585 and X555 discrete driver does not support the ink slider feature.

1. Open the printer driver dialog box, select the color tab, and then do the following:
 - a. Uncheck the HP EasyColor check box.
 - b. Select the Ink Settings button.
ACCESS THE INK SLIDER



2. The following settings can be configured using the ink slider:
 - Dry Time: Use this item to adjust the print speed (default is 0). Move the slider to the right to slow down the print speed.
 - Saturation: Use this item to adjust the amount of ink used (default is 0). Move the slider to the right (1 and 2) to increase the amount of ink used. Move the slider to the left (-1 and -2) to decrease the amount of ink used.
 - Select the Black Ink Spread: Use this item to improve bleed through when black and other colors are printing in certain patterns.



Advanced cleaning

Use the following procedure to perform Advanced cleaning.

The advance cleaning feature attempts to remove any fiber or contaminants stuck to the shim underneath print bar.

NOTE:

To use the advanced cleaning feature, order the advanced cleaning kit (part number CN459-67006).

1. At the printer control panel, open the following menus:
 - Device maintenance
 - Calibration/Cleaning
 - Advanced Cleaning
2. Follow the instruction provided in the advanced cleaning kit (CN459-67006) to install the kit in the tray, and to perform the advanced cleaning procedure.



View a VIDEO with the steps to use the Advanced Cleaning kit, Go to the following link : [Clean the Print Bar Using the Advanced Cleaning Kit | HP Printer | HP](#)